

KERALA POLYTECHNIC SYLLABUS • REV2021

Engineering Chemistry Lab

Course Code: 2079

Semester: 2

Credits: No Credit

Dept: Chemical / Food Tech

[Official Syllabus PDF](#)

Official Source Declaration

This lesson is architected based on the official **SITTTR Kerala REV2021** syllabus for Course Code **2079 (Engineering Chemistry Lab)**. All modules, radicals, and assessment criteria are derived directly from the authoritative document provided for Diploma in Chemical Engineering and Food Processing Technology.

Note: While the syllabus lists "No Credit", the course involves 45 contact hours and two internal Lab Examinations (Lab Exam I & II) which are critical for semester progression.

Course Overview

Teaching Scheme

- **L:T:P** — 3:0:0 (Practical oriented)
- **Total Periods** — 45 per semester
- **Category** — Engineering Science

Course Objectives

To provide hands-on experience in the qualitative analysis of Inorganic compounds as simple salts and binary mixtures.

Malayalam support:

ഈ കോഴ്സ് വിദ്യാർത്ഥികൾക്ക് അനോറിക് സംയുക്തങ്ങൾ (Inorganic compounds) ക്വലിറ്റേറ്റീവ് അനലിസിസ് (Qualitative Analysis) ചെയ്യാനുള്ള അറിവ് നൽകുന്നു. അനോറിക് സംയുക്തങ്ങൾ (Simple Salts) അല്ലെങ്കിൽ ദ്വയാർത്ഥികൾ (Binary Mixtures) ക്വലിറ്റേറ്റീവ് അനലിസിസ് ചെയ്യാനുള്ള അറിവ് നൽകുന്നു.

Course Outcomes (CO)

CO No.	Description	Duration	Cognitive Level
CO1	Analyse acid radicals	9 Hours	Applying
CO2	Analyse basic radicals	10 Hours	Applying
CO3	Analyse simple salts	12 Hours	Applying

CO No.	Description	Duration	Cognitive Level
CO4	Analyse binary mixtures	14 Hours	Applying

Module 1: Analysis of Acid Radicals (Anions)

9 Hours

Acid radicals are the negatively charged ions in an inorganic salt. The analysis involves systematic identification using preliminary and confirmatory tests.

Scope of Radicals

The syllabus requires identification of the following acid radicals:

- Carbonate (CO_3^{2-})
- Sulphide (S^{2-})
- Sulphite (SO_3^{2-})
- Sulphate (SO_4^{2-})
- Thiosulphate ($\text{S}_2\text{O}_3^{2-}$)
- Nitrate (NO_3^-)
- Fluoride (F^-)
- Chloride (Cl^-)
- Bromide (Br^-)
- Iodide (I^-)
- Acetate (CH_3COO^-)
- Oxalate ($\text{C}_2\text{O}_4^{2-}$)
- Phosphate (PO_4^{3-})
- Borate (BO_3^{3-})
- Chromate (CrO_4^{2-})

Systematic Procedure for Acid Radicals

Module 2: Analysis of Basic Radicals (Cations)

10 Hours

Basic radicals are positively charged ions. They are classified into groups based on their solubility products and identified using group reagents.

Group Classification

Group	Radicals	Group Reagent
Group 0	Ammonium (NH_4^+)	NaOH
Group I	Silver (Ag^+), Lead (Pb^{2+}), Mercury (Hg_2^{2+})	Dilute HCl
Group II	Copper (Cu^{2+}), Cadmium (Cd^{2+}), Bismuth (Bi^{3+}), Arsenic (As^{3+}), Antimony (Sb^{3+}), Tin ($\text{Sn}^{2+}/\text{Sn}^{4+}$)	H_2S gas in presence of Dil. HCl
Group III	Iron (Fe^{3+}), Aluminium (Al^{3+}), Chromium (Cr^{3+})	NH_4OH in presence of NH_4Cl
Group IV	Zinc (Zn^{2+}), Manganese (Mn^{2+}), Cobalt (Co^{2+}), Nickel (Ni^{2+})	H_2S gas in presence of NH_4OH

Group	Radicals	Group Reagent
Group V	Barium (Ba^{2+}), Strontium (Sr^{2+}), Calcium (Ca^{2+})	$(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4Cl & NH_4OH
Group VI	Magnesium (Mg^{2+}), Sodium (Na^+), Potassium (K^+)	Specific reagents (e.g., Na_2HPO_4 for Mg)

Module 3: Analysis of Simple Salts

12 Hours

A simple salt consists of one acid radical and one basic radical. The analysis must follow a systematic workflow:

1. Preliminary Examination (Color, State, Solubility).
2. Identification of Acid Radical.
3. Confirmatory test for Acid Radical.
4. Preparation of Original Solution (OS).
5. Identification of Basic Radical Group.
6. Identification of Basic Radical within the group.
7. Confirmatory test for Basic Radical.

Syllabus Constraint: The simple salt should be soluble in water, dilute acids, or conc. HCl and should not involve interfering radicals (like Fluoride, Borate, Oxalate, Phosphate) that complicate group separation.

Module 4: Analysis of Binary Mixtures

14 Hours

Binary mixtures contain two simple salts, meaning two acid radicals and two basic radicals must be identified.

Strategy: In binary mixtures, one must check if the presence of one radical interferes with the test for another. For example, in a mixture of Chloride and Bromide, special separation is needed.

Open Ended Experiments (3 Hours)

Students are encouraged to apply their analytical skills to unknown samples or real-world materials like water hardness or soil minerals (qualitative aspect).

Formula Bank & Calculations

Solubility Product (K_{sp})

The principle behind group separation. A precipitate forms when the Ionic Product exceeds K_{sp} .

Concept: For a salt A_xB_y ,
 $K_{sp} = [A]^x [B]^y$

Common Ion Effect

Used to suppress the ionization of weak electrolytes. Example: NH_4Cl suppresses NH_4OH ionization to ensure only Group III hydroxides precipitate.

Molarity Calculator (For Reagent Prep)

Mass of Solute (g)

e.g. 40

Molar Mass (g/mol)

e.g. 40 for NaOH

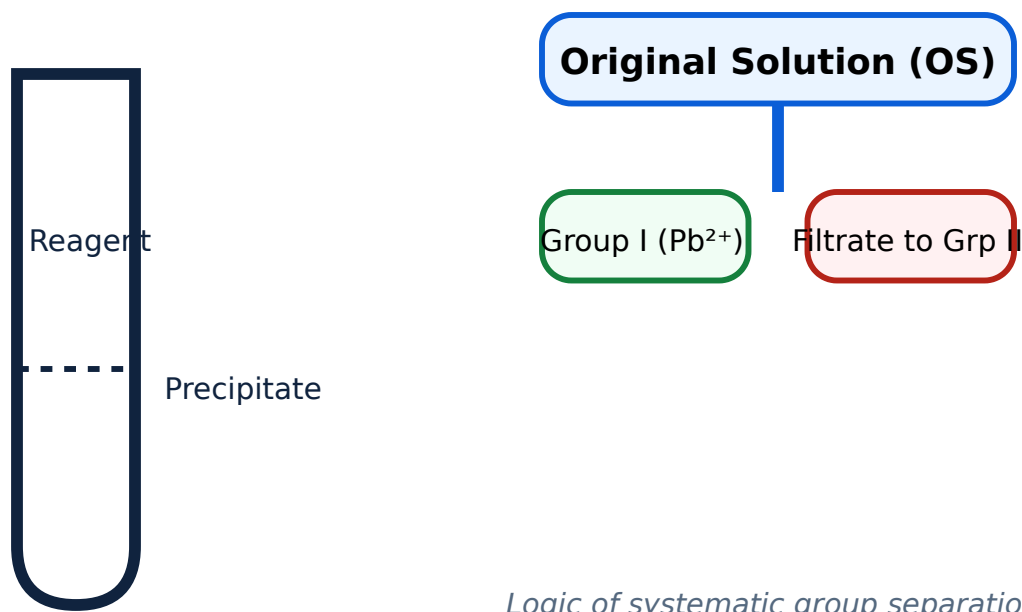
Volume of Solution (mL)

e.g. 1000

Calculate Molarity

Result: -- M

Visual Library



Basic setup for qualitative precipitation tests.

Logic of systematic group separation.

Practice Questions & Answers

- 1. Why is NH_4Cl added before NH_4OH in Group III analysis?

► 2. How do you identify the presence of Carbonate radical?

► 3. What is the confirmatory test for Nitrate?

Practice Model Paper

PRACTICE MODEL EXAM — NOT AN OFFICIAL PAPER

Time: 3 Hours | Max Marks: 100 (Internal Pattern)

Part A: Preliminary (20 Marks)

1. Perform physical examination and solubility test for the given salt mixture.
2. Report the color, state, and solubility in different solvents.

Part B: Acid Radical Analysis (30 Marks)

3. Identify and confirm two acid radicals present in the given binary mixture with systematic steps.

Part C: Basic Radical Analysis (40 Marks)

4. Identify and confirm two basic radicals present in the given binary mixture using group separation.

Part D: Viva & Record (10 Marks)

5. Answer questions on group reagents and solubility products.

Quick Revision

Group Reagents

- Gr I: Dil. HCl
- Gr II: $\text{H}_2\text{S} + \text{HCl}$
- Gr III: $\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$

Flame Test Colors

- Barium: Apple Green
- Strontium: Crimson Red
- Calcium: Brick Red

Common Mistakes

- Not using freshly prepared FeSO_4 for Nitrate test.
- Adding too much NH_4OH in Group III.

Glossary

- **Qualitative Analysis:** Identification of the constituents of a substance.
- **Precipitate:** An insoluble solid that emerges from a liquid solution.
- **Original Solution (OS):** A clear solution of the salt in a suitable solvent (Water/HCl).
- **Radical:** An atom or group of atoms carrying a charge.

References

- Practical Chemistry for B.Sc Main - A.O. Thomas and Mani.
- NCERT Laboratory Manual - Chemistry (Class XI & XII).
- SITTTTR Kerala Official Syllabus Revision 2021.

മലയാളം പിന്തുണ (Malayalam Support)

ഈ പാഠ്യപുസ്തകത്തിൽ ഉൾക്കൊള്ളിച്ചിരിക്കുന്ന വിവരങ്ങൾ:

- **Original Solution (OS):** പരീക്ഷിക്കേണ്ടുന്ന ലവണത്തിന്റെ വ്യക്തമായ ലായനം. ലായനം വ്യക്തമായതും അലിഞ്ഞു കലർന്നതും ആയിരിക്കണം.
- **Group Separation:** ലവണങ്ങളെ ഗ്രൂപ്പുകളായി വേർതിരിക്കുന്ന പ്രക്രിയ. ലവണങ്ങളെ വേർതിരിക്കുന്നതിനായി ഉപയോഗിക്കുന്ന രാസവസ്തുക്കൾ.

- **Confirmatory Test:** ພິມ ພິມພິມພິມ ພິມພິມພິມ ພິມພິມພິມພິມ ພິມພິມພິມ ພິມພິມພິມ ພິມພິມພິມ ພິມພິມພິມພິມ ພິມພິມ ພິມພິມພິມ.